

Magnetism and Moving Charges

(Class 12th SURU VALLEY PUBLIC HIGHER SECONDARY SCHOOL
KARGIL)

- Q1.** A proton enters a magnetic field of 0.5 T with a speed of 2×10^6 m/s perpendicular to the field. Calculate the magnetic force acting on it.
- Q2.** An electron moves with a speed of 4×10^6 m/s making an angle of 30° with a magnetic field of 0.25 T. Find the magnetic force.
- Q3.** A particle carrying a charge of $5 \mu\text{C}$ moves parallel to a magnetic field of 0.8 T. Find the magnetic force.
- Q4.** A charge of $8 \mu\text{C}$ experiences a magnetic force of 0.16 N while moving perpendicular to a magnetic field of 0.4 T. Find its speed.
- Q5.** An alpha particle enters a magnetic field normally with speed 3×10^6 m/s. If the field is doubled, what happens to the magnetic force?
- Q6.** A charge moving at 5×10^5 m/s experiences a force of 2×10^{-4} N in a magnetic field of 0.2 T. Find the charge.
- Q7.** An electron enters a magnetic field of 0.1 T perpendicular to it with a speed of 4×10^6 m/s. Find the radius of its circular path.
- Q8.** A proton moves perpendicular to a magnetic field of 0.4 T. Calculate the time period of revolution.
- Q9.** Find the cyclotron frequency of a proton in a magnetic field of 0.8 T.
- Q10.** Calculate the angular frequency of an electron moving in a magnetic field of 0.2 T.
- Q11.** A proton has velocity components 3×10^6 m/s parallel and 2×10^6 m/s perpendicular to a magnetic field of 0.5 T. Calculate the pitch of the helical path.
- Q12.** A particle moves in a circle of radius 0.12 m inside a magnetic field of 0.6 T. Its speed is 3×10^6 m/s. Find the charge-to-mass ratio.
- Q13.** A wire 40 cm long carries a current of 5 A and is placed perpendicular to a magnetic field of 0.6 T. Find the force.
- Q14.** A 1 m conductor carrying 8 A makes an angle of 60° with a magnetic field of 0.5 T. Find the force.
- Q15.** A wire experiences a force of 3 N in a field of 0.75 T. The current is 10 A. Find its length.
- Q16.** A conductor of length 50 cm carrying 6 A experiences a force of 0.9 N. Find the magnetic field.

- Q17.** A current-carrying conductor is placed parallel to magnetic field lines. What force acts on it?
- Q18.** Two long parallel wires 10 cm apart carry currents of 6 A and 8 A in the same direction. Find the force per metre between them.
- Q19.** Two wires 20 cm apart carry currents of 12 A and 15 A in opposite directions. Calculate the force per metre.
- Q20.** Find the current in each of two wires separated by 1 m if the force per metre between them is 2×10^{-7} N/m.
- Q21.** Calculate the magnetic field at a point 5 cm from a long straight wire carrying 10 A.
- Q22.** A wire carries 20 A. Find the magnetic field at a distance of 8 cm.
- Q23.** At what distance from a straight conductor carrying 12 A will the magnetic field be 4×10^{-5} T?
- Q24.** A current element of length 4 mm carries 6 A. Find the magnetic field at a point 20 cm away when the element is perpendicular to the radius vector.
- Q25.** A long straight conductor carries 25 A. Calculate the magnetic field 10 cm away.
- Q26.** A circular coil of radius 10 cm carries 5 A. Find the magnetic field at its centre.
- Q27.** A circular coil has 50 turns, radius 8 cm, and carries 3 A. Calculate the magnetic field at the centre.
- Q28.** A long solenoid has 1200 turns over a length of 60 cm. If it carries 2 A, find the magnetic field inside it.
- Q29.** A solenoid has 1500 turns over 75 cm and carries 4 A. Find the magnetic field inside.
- Q30.** A toroid has 800 turns carrying 5 A. The mean radius is 12 cm. Find the magnetic field inside the toroid.
- Q31.** A toroid has 500 turns, current 3 A, and mean radius 8 cm. Find the magnetic field.
- Q32.** A circular coil of 40 turns, radius 5 cm, carries 2 A. Find its magnetic dipole moment.
- Q33.** A rectangular coil of dimensions 20 cm $\times$ 10 cm has 30 turns carrying 4 A. Calculate its magnetic dipole moment.
- Q34.** A coil has magnetic moment 0.8 A m^2 placed in a magnetic field of 0.5 T at 30° . Find the torque.
- Q35.** A coil has $N = 100$, area $A = 0.02 \text{ m}^2$, current $I = 0.5 \text{ A}$, and is placed perpendicular to a magnetic field of 0.3 T. Calculate the maximum torque.
- Q36.** A bar magnet has pole strength 4 A m and magnetic length 8 cm. Find its magnetic dipole moment.
- Q37.** A magnetic dipole of moment 1.5 A m^2 is placed parallel to a magnetic field of 0.4 T. Find its potential energy.

- Q38.** A magnetic dipole of moment 0.6 A m^2 is placed at 60° to a magnetic field of 0.8 T . Find the potential energy.
- Q39.** A bar magnet has magnetic moment 5 A m^2 . Find the magnetic field at a point 20 cm from its centre on the axial line.
- Q40.** A bar magnet has magnetic moment 3 A m^2 . Find the magnetic field at a point 15 cm away on the equatorial line.

Useful Constants

- Permeability of free space:

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

- Charge of electron:

$$e = 1.6 \times 10^{-19} \text{ C}$$

- Mass of electron:

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

- Mass of proton:

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

-

$$\pi = 3.14$$

Enjoy your vacation and **Email the assignment to** ratherjunaid67@gmail.com